Blinkit Sales Analytics Report

Problem Statement

Blinkit is an online grocery delivery platform that handles thousands of customer orders across different cities, products, and payment methods. However, understanding sales performance, customer behavior, delivery efficiency, and product demand can be difficult using raw data.

The goal of this project is to analyse Blinkit sales data using Power BI to identify business trends, customer preferences, top-performing products, delivery issues, and payment patterns. The dashboard helps convert raw data into meaningful insights for better business decision-making.

Dataset Information

The dataset contains information about **orders, customers, products, sales, and transactions**. It includes :

- **Order Details :** Order ID, Date, Delivery
- **Customer Details :** ID, Gender, Age Group, Membership
- **Location Details :** State, City, Region
- **Product Details :** Name, Category, Brand
- **Sales Metrics :** Price, Quantity, Sales, Profit, Discount)
- **Transaction Details :** Payment Method, Rating, Return Status.

Data Preprocessing

- Removed duplicate records
- Handled null/blank values
- Converted date columns into proper format
- Verified numerical data types
- Standardized column names
- Checked inconsistent category and city names
- Cleaned and formatted data for analysis
- Created KPIs like **Profit Margin** and **Average Order Value**

All Chart Explanations

A) Sales Overview

1.1 Total Sales by Order_Date (Line Chart)

This chart shows how total sales changed over time based on order dates.
Insight : Your sales trend is gradually decreasing, meaning sales performance reduced over time.

1.2 Total Profit by Payment Method (Pie Chart)

It tells us which payment method contributes the highest profit.
Insight : UPI contributes the highest profit share, meaning customers prefer digital payments.

1.3 Sum of Total Amount by Category (Bar Chart)

This chart compares sales revenue across product categories.
Insight : Top categories generate around 0.68M–0.71M sales, showing strong demand.

1.4 Sum of Customer Rating by Payment Method (Bar Chart)

We can see which payment method customers prefer and rate more positively.
Insight : UPI has the highest rating value, meaning customers using UPI appear more satisfied.

1.5 Total Profit by State (Area Chart)
This chart displays profit generated from each state.
Insight : Maharashtra generates the highest profit, while Delhi has lower profit.

B) Product / Item Analysis
2.1 Total Sales by Category (Bar Chart)
This chart displays total sales generated by each product category.
Insight : Snacks and Fruits & Vegetables generate the highest sales, while categories like Personal Care and Dairy contribute comparatively less.

2.2 Total Sales by Brand (Treemap)
Treemap helps visualize **which brands contribute more revenue** using box size.
Insight : Amul and Britannia are top-performing brands, contributing the highest sales.

2.3 Total Sales by Product Name (Bar Chart)
To identify best-selling products.
Insight : Frozen Peas is the highest-selling product, followed by Cleaner and Detergent.

2.4 Sum of Item_MRP and Selling Price by Product Name (Scatter Chart)
This chart shows the relationship between MRP and Selling Price.
Insight : There is a positive relationship between MRP and Selling Price, meaning as product MRP increases, selling price also increases.

2.5 Discount Percentage and Total Sales (Donut Chart)
To understand the **impact of discounts on overall sales**.
Insight : Total Sales contribute nearly **99%**, while average discount is around **20%**.

C) Customer & Delivery Analysis
3.1 Delivery Status Distribution (Donut Chart)
A donut chart helps understand the percentage share of each delivery status.
Insight : The distribution is almost balanced across statuses, but Delivered orders have a strong share.
However, Cancelled and Returned orders are also high, which may indicate operational issues.

3.2 Average Delivery Time by City (Bar Chart)
This chart compares **average delivery time across cities**.
Bar charts help compare delivery performance between cities.
Insight : Mysore has the highest delivery time (~68 minutes), while Pune and Kolkata have comparatively faster delivery.

3.3 Customer Rating Analysis (Column Chart)
To understand customer satisfaction compared with order volume.
Insight : Customer ratings are moderate, which matches the KPI average rating (2.99).

3.4 Gender-wise Orders (Pie Chart)

This chart compares orders placed by customer gender.

Insight : Orders are fairly balanced, but female customers contribute slightly more orders.

3.5 Cancelled Orders by City (Map Chart)

This map displays **city-wise cancelled orders**.

Insight : Some cities show **higher cancellation concentration**, meaning operational challenges exist in those locations.

D) Business Insights & Trends

4.1 Total Sales by Order_Date (Line Chart)

This chart displays sales trends over time based on order dates.

Insight : Sales show continuous fluctuations with several peaks and drops instead of stable growth.

4.2 Total Amount by Payment Method (Bar Chart)

This chart compares sales revenue generated through each payment method.

Insight : UPI generates the highest revenue, much higher than other payment methods.

4.3 Product Stock Availability (Treemap)

This chart displays product stock availability status.

Insight : Most products are In Stock, while some products are Limited or Out of Stock.

4.4 Total Amount by City (Map Chart)

This map visualizes sales distribution across cities.

Insight : Sales are spread across multiple cities, but some cities contribute more sales than others.

4.5 Quantity by Category (Donut Chart)

Donut charts help understand category contribution percentage.

Insight : Sales quantity is fairly balanced across categories, but Frozen Food and Beverages have slightly higher contribution.

DAX Measures Used

1. Delivered Orders =

```
CALCULATE(
    COUNT('blinkit_dataset'[Order_ID]),
    'blinkit_dataset'[Order_Status] = "Delivered"
)
```

2. Cancelled Orders =

```
CALCULATE(
    COUNT('blinkit_dataset'[Order_ID]),
    'blinkit_dataset'[Order_Status] = "Cancelled"
)
```

3. Average Discount =

```
AVERAGE('blinkit_dataset'[Discount_Percentage])
```

4. Average Delivery Time =

```
AVERAGE('blinkit_dataset'[Delivery_Time_Minutes])
```

5. ```
   Total Customers =
   DISTINCTCOUNT(blinkit_dataset[Customer_ID])
   ```

Key Insights

1. Strong Overall Revenue Performance

- Blinkit generated approximately ₹6M in total sales from 5K orders.

2. Customer Spending Pattern

- The average sale value is ₹1.18K per order.

3. Customer Satisfaction Needs Improvement

- The average customer rating is 2.99, which is below ideal.
- This indicates issues related to:
  - delivery delays
  - product quality
  - customer service

4. UPI is the Most Preferred Payment Method

5. High Cancellation Rate is a Major Concern

6. Maharashtra generated the highest profit and customer ratings.

Recommendations

1. Improve customer service to increase customer ratings.
2. Reduce cancelled orders by improving delivery and stock management.
3. Make delivery faster in cities with high delivery time.
4. Keep more stock of high-selling products and categories.
5. Give cashback and offers on UPI payments to attract customers.
6. Focus more on high-performing states like Maharashtra.

Final Conclusion

This Blinkit project helped analyze sales, customers, products, and delivery performance. The dashboard provides useful insights to improve sales, customer satisfaction, and business growth.

END OF REPORT
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